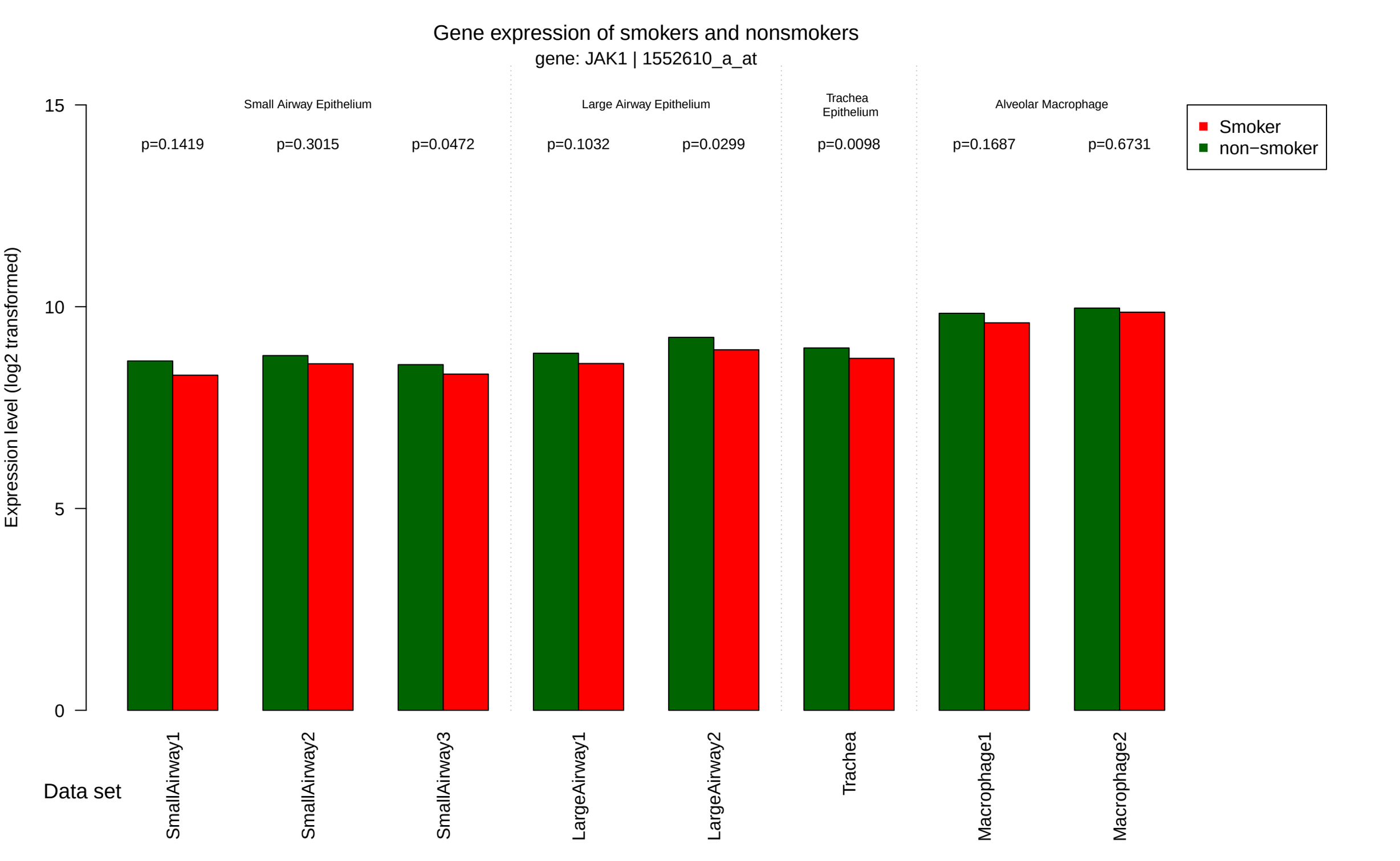
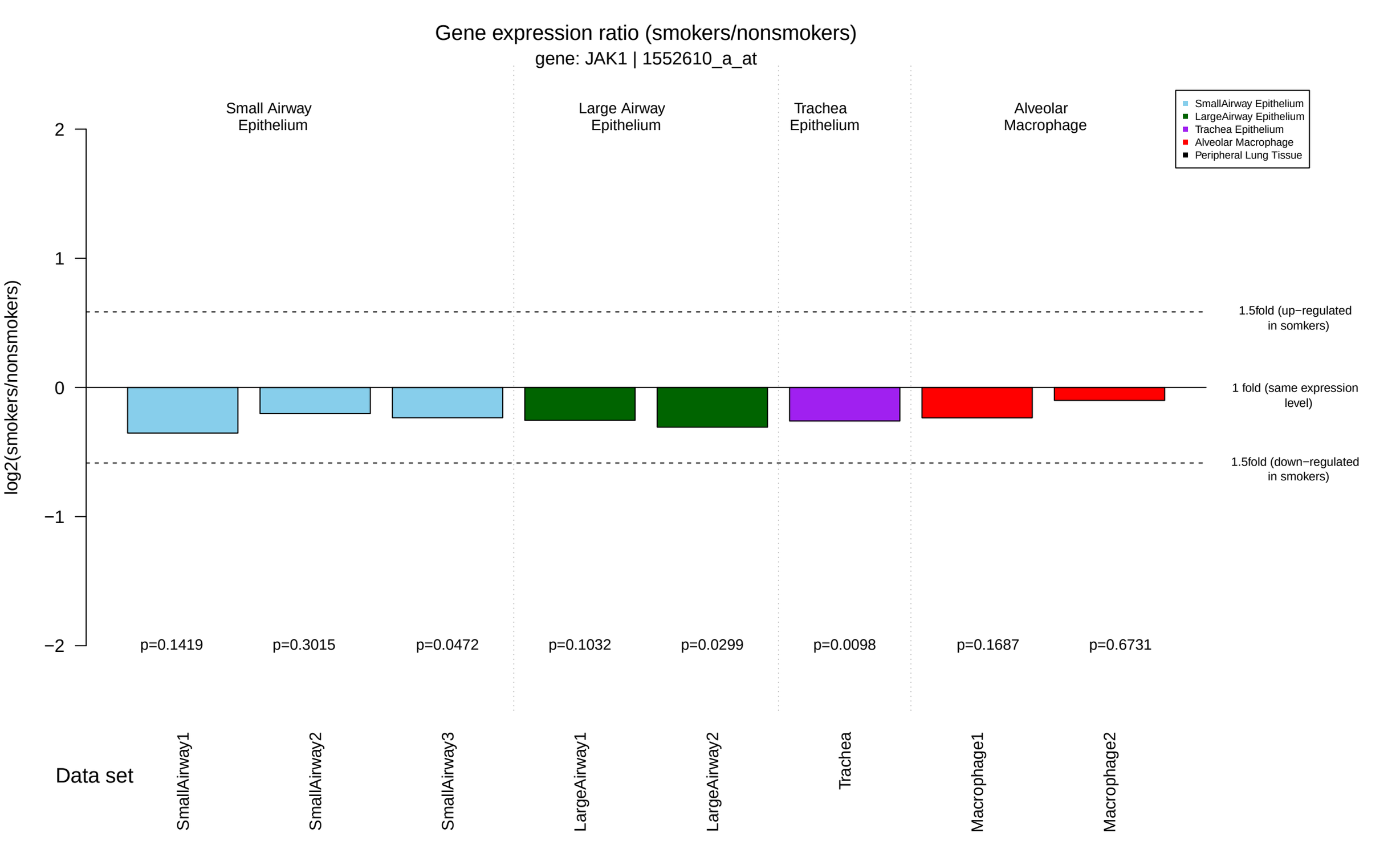
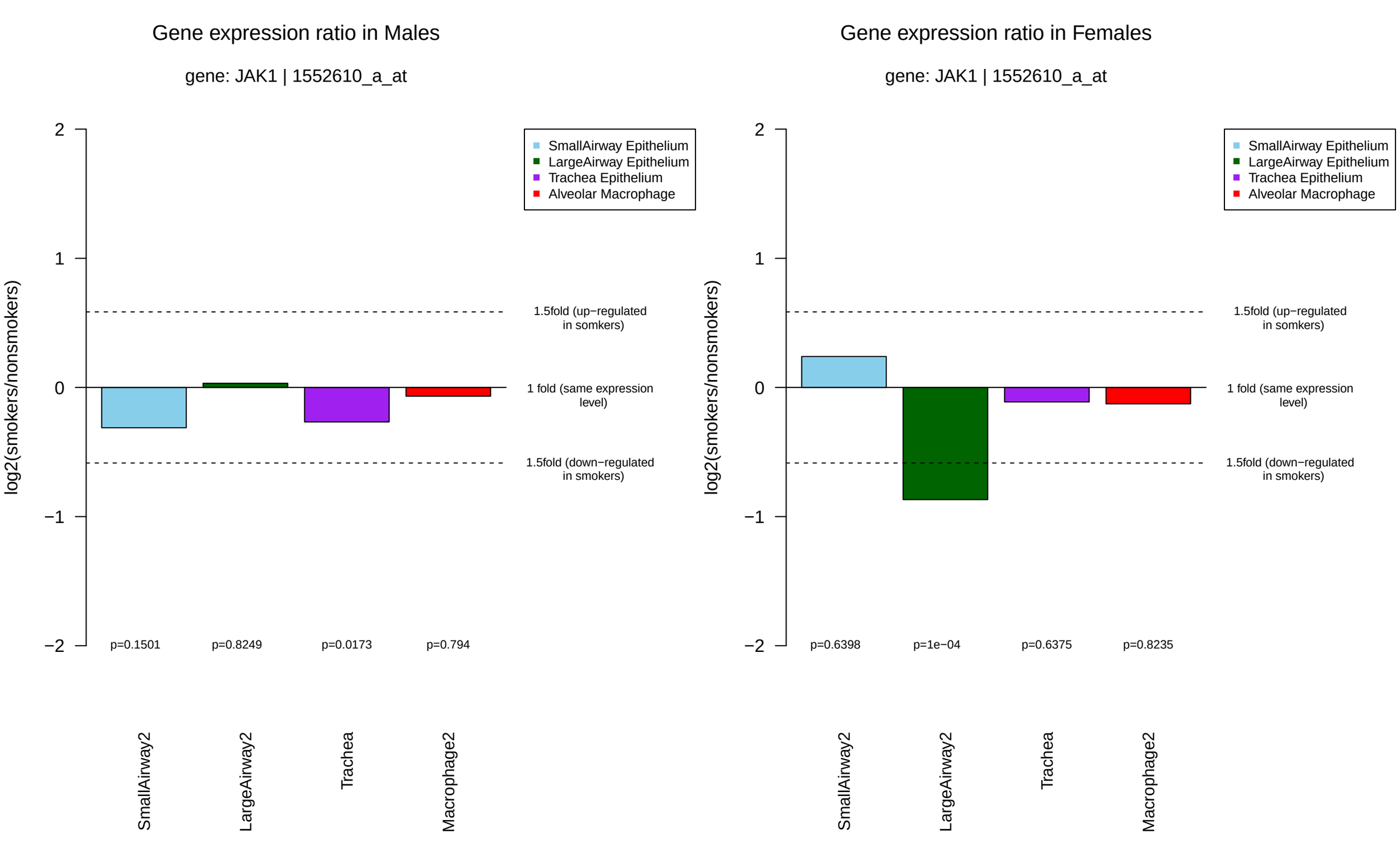
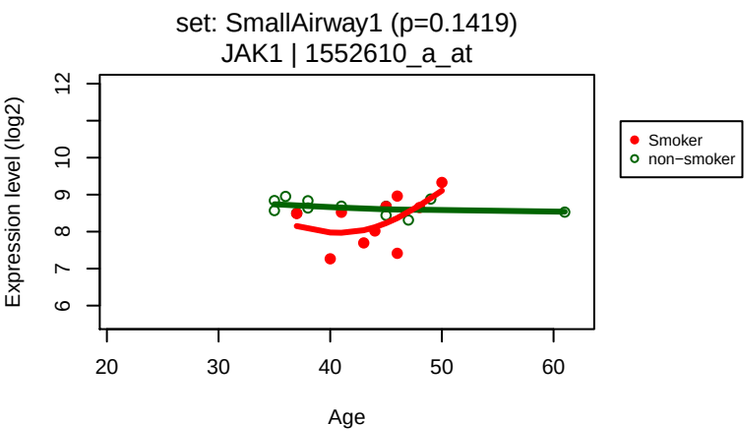
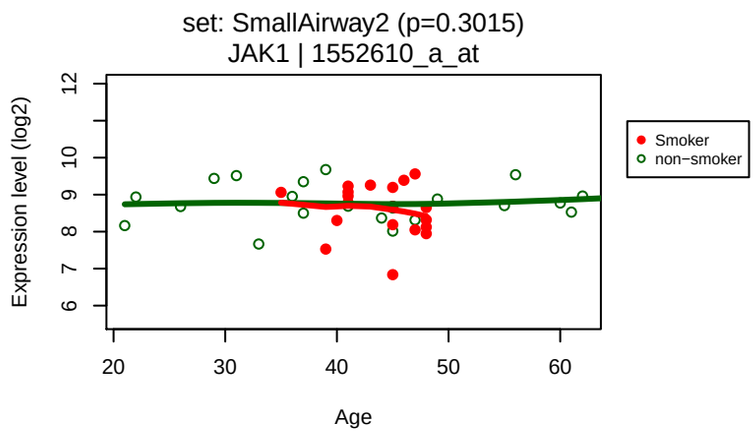
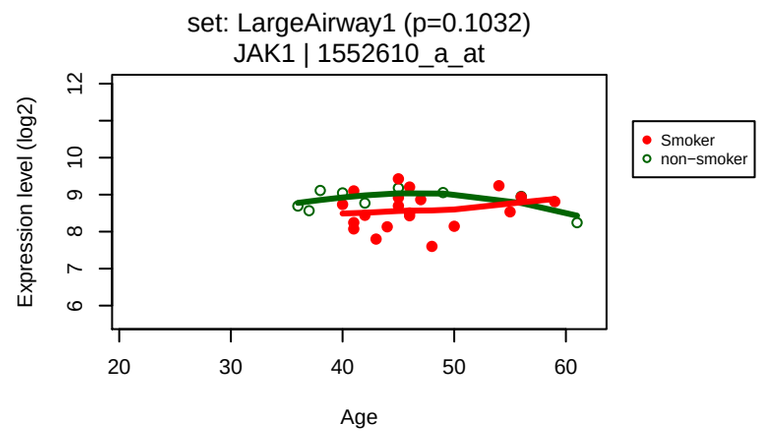

Gene expression of smokers and nonsmokers
gene: JAK1 | 1552610_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1419
p=0.3015
p=0.0472
p=0.1032
p=0.0299
p=0.0098
p=0.1687
p=0.6731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

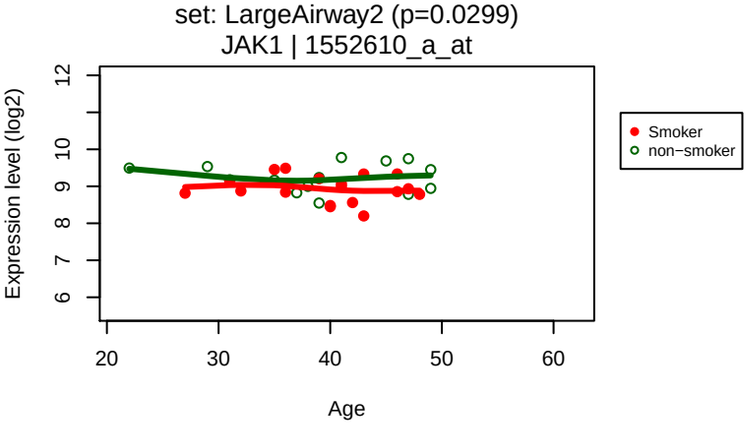

Gene expression ratio (smokers/nonsmokers)
gene: JAK1 | 1552610_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1419
p=0.3015
p=0.0472
p=0.1032
p=0.0299
p=0.0098
p=0.1687
p=0.6731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

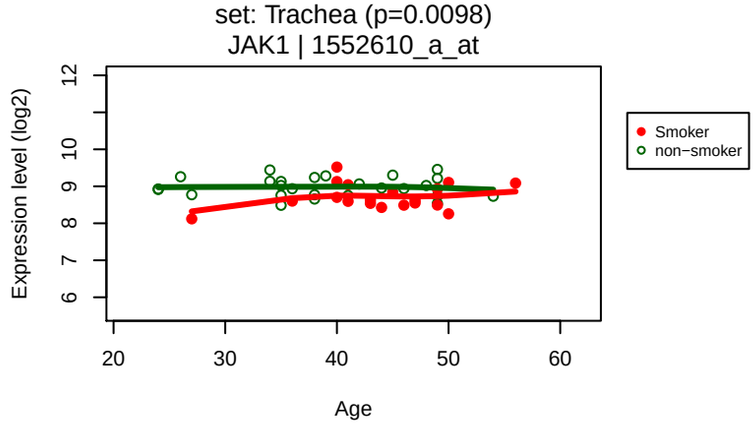

Gene expression ratio in Males
gene: JAK1 | 1552610_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1501
p=0.8249
p=0.0173
p=0.794
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK1 | 1552610_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6398
p=1e−04
p=0.6375
p=0.8235
SmallAirway2
LargeAirway2
Trachea
Macrophage2

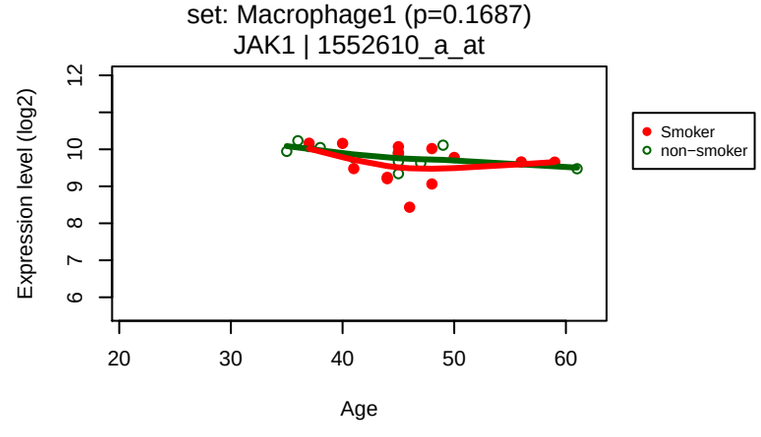

set: SmallAirway1 (p=0.1419)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker

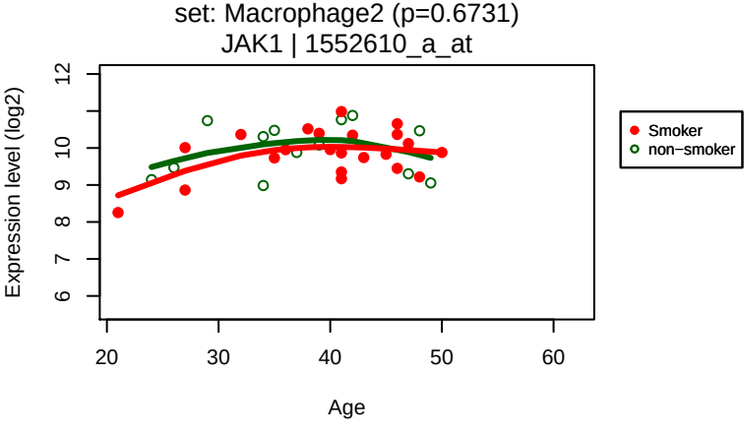

set: SmallAirway2 (p=0.3015)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker

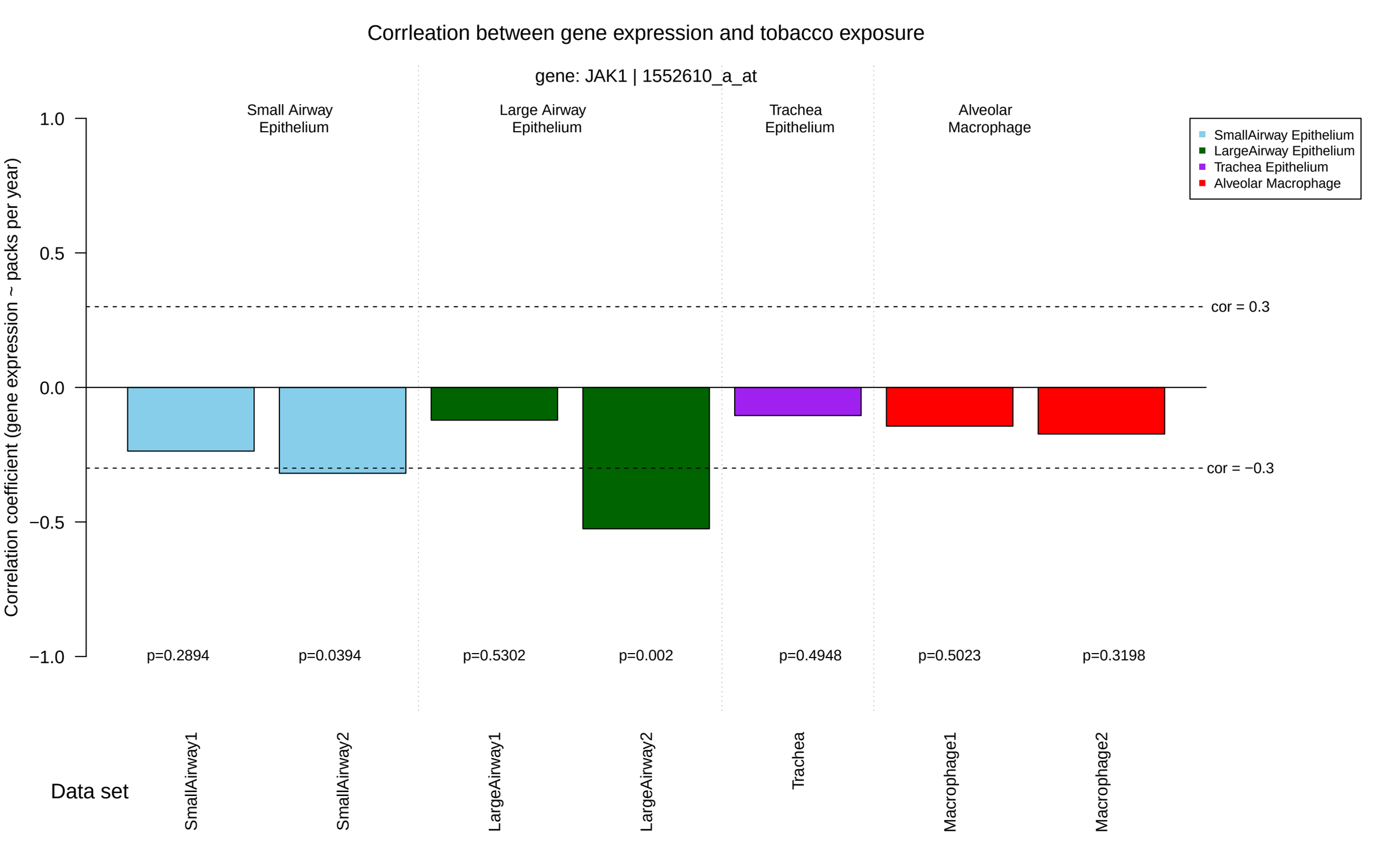

set: LargeAirway1 (p=0.1032)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0299)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0098)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1687)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6731)
JAK1 | 1552610_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK1 | 1552610_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2894
p=0.0394
p=0.5302
p=0.002
p=0.4948
p=0.5023
p=0.3198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2